THEORETICAL & APPLIED MECHANICS LETTERS **4**, 021007 (2014)

Failure criteria for C/SiC composites under plane stress state

Chengpeng Yang,[a)] Guiqiong Jiao, Hongbao Guo
Department of Engineering Mechanics, Northwestern Polytechnical University, Xi'an 710129, China

(Received 30 November 2013; revised 13 February 2014; accepted 20 February 2014)

Abstract The applicability and limitation of several quadratic strength theories were investigated with respect to 2D-C/SiC and 2.5D-C/SiC composites. A kind of damage-based failure criterion, referred to as D-criterion, is proposed for nonlinear ceramic composites. Meanwhile, the newly developed criterion is preliminarily validated under tension-shear combined loadings. The prediction shows that the failure envelope given by D-criterion is lower than that from Tsai–Hill and Hoffman criteria. This reveals that the damage-based criterion is reasonable for evaluation of damage-dominated failure strength.
 [doi:10.1063/2.1402107]

Keywords C/SiC composite, damage evolution, failure criterion

Few studies have been dedicated to the failure criterion for ceramic matrix composites due to the material nonlinearity, anisotropy and inhomogeneity, as well as the complexity of the directional interior microstructural damages. In this study, the applicability and limitation of several quadratic strength theories are investigated with respect to C/SiC composites. To overcome the drawbacks, a new failure criterion based on damage evolution mechanisms is developed for C/SiC composites and/or other nonlinear brittle matrix composites.

Quadratic failure theories such as Tsai–Hill criterion,[1] Hoffman criterion,[2] and Tsai–Wu tensorial criterion[3] are widely accepted for orthotropic materials. However, these phenomenological theories are obviously based on data-fitting approach, without any physical significance. In the present analysis, we encounter highly anisotropic failure problems, and a Cartesian material reference system is considered under plane stress states. We focus on the failure assessment in two fracture planes, one perpendicular to axis-1 (referred to as plane-1) and the other perpendicular to axis-2 (referred to as plane-2).

The Tsai–Hill criterion was formulated by referring to distortional energy and is thus an interactive criterion. If the longitudinal stress σ_1 and shear stress τ_{12} are imposed on plane-1, the failure condition can be given by the following equation

$$(\sigma_1/X)^2 + (\tau_{12}/S_{12})^2 = 1, \tag{1a}$$

where X is the allowable value of σ_1 and S_{12} is the positive pure shear strength. Similarly, if one impose the transverse stress σ_2 and shear stress τ_{21} on plane-2, the failure condition is

$$(\sigma_2/Y)^2 + (\tau_{21}/S_{21})^2 = 1, \tag{1b}$$

[a)]Corresponding author. Email: yang@mail.nwpu.edu.cn.

where Y is the maximum value of σ_2 and S_{21} is the negative pure shear strength. It is considered that the Tsai–Hill criterion fails to account for the different strength behavior in tension and compression of the anisotropic materials. In this aspect, Hoffman made some improvements by incorporating linear terms into the fracture condition.

Under plane stress state, the Hoffman criterion for combined loading of longitudinal stress and shear stress on plane-1 can be stated as

$$\sigma_1^2/(X_tX_c) + (X_c - X_t)/(X_tX_c)\sigma_1 + \tau_{12}^2/S_{12}^2 = 1, \tag{2a}$$

where X_t and X_c are the longitudinal tensile and compressive strength, respectively. While for the similar combined loadings on plane-2, the failure condition can be expressed as

$$\sigma_2^2/(Y_tY_c) + \sigma_2(Y_c - Y_t)/(Y_tY_c) + \tau_{21}^2/S_{21}^2 = 1, \tag{2b}$$

where Y_t and Y_c are the transverse tensile and compressive strength, respectively. Once the tensile strength is equal to the compressive strength in each principal material direction (i.e., axis-1 and axis-2), Eqs. (2a) and (2b) will revert to Eqs. (1a) and (1b), respectively.

The failure criterion of Tsai–Wu is conceived, different from Tsai–Hill and Hoffman, for an entirely anisotropic material. Consequently, all of the quadratic interaction terms about normal stresses and shear stress are included in the scalar formulation. When a state of plane stress is applied to plane-1 or plane-2, the Tsai–Wu failure condition can be formulated as

$$F_{ij}\sigma_i\sigma_j + F_i\sigma_i = 1, \quad i,j = 1,2,6, \tag{3}$$

where σ_6 is the in-plane shear stress (τ_{12} or τ_{21}). The components of F_{ij} and F_i are related to the strength properties of the material, X_t, X_c, Y_t, Y_c, S_{12}, S_{21}. If mechanical rupture initiates in plane-1 by tension-shear or compression-shear combinations, the following condition can be used for failure prediction

$$\sigma_1^2/(X_tX_c) + \sigma_1(X_c - X_t)/(X_tX_c) + \tau_{12}(S_{21} - S_{12})/(S_{12}S_{21}) + \tau_{12}^2/(S_{12}S_{21}) = 1. \tag{4a}$$

However, if the fracture originates in plane-2 due to similar combinations, the simplified Tsai–Wu condition showed below can be suggested

$$\sigma_2^2/(Y_tY_c) + \sigma_2(Y_c - Y_t)/(Y_tY_c) + \tau_{21}(S_{12} - S_{21})/(S_{12}S_{21}) + \tau_{21}^2/(S_{12}S_{21}) = 1. \tag{4b}$$

Note that in the case of $S_{12} = S_{21}$, Eqs. (4a) and (4b) will revert to Eqs.(2a) and (2b), respectively.

In order to validate the applicability of the noted strength theories described previously, an examination is performed by comparing the theoretical predictions with experimental data listed in literatures.

For 2D-C/SiC composites with weak fiber/matrix interface, the average measured data under tension-shear combination loadings[4] are depicted in Fig. 1. The reported compression strength is $X_c = 430.2$ MPa. It can be seen from this figure that the distribution of the data is mainly in elliptical shape. Therefore, the quadratic criteria may be appropriate for curve fitting. But obviously, the Tsai–Hill criterion overestimates the bearing capacity of the material to some extent, whereas the failure envelope simulated by Tsai–Wu and/or Hoffman is relatively in better agreement with the experimental data.

According to the test results, Guan et al.[4] suggested a new quadratic equation to describe the strength character of 2D-C/SiC composites. The criterion takes into account the inequality between tensile strength and compressive strength, and shear stress recovery due to crack closure caused by compression. Furthermore, the failure condition given in Ref. 4 can be transformed into a more complicated equation for plane stress state as

$$A\left(\sigma_1^2+\sigma_2^2\right)-2A\left(\sigma_1+\sigma_2\right)\sigma_0+A\sigma_0^2+B\tau_{12}^2=1. \tag{5}$$

The coefficients A, B, and σ_0 can be determined by

$$A=4/(X_t+X_c)^2,\ \ \sigma_0=(X_t-X_c)/2,\ \ B=[4X_tX_c/(X_t+X_c)^2](1/S_{12}^2). \tag{6}$$

Note that Eq. (5) is acceptable for brittle composites that present the same mechanical properties along each principal material direction, i.e., $X_t=Y_t$, $X_c=Y_c$. The estimated result of Eq. (5) is very close to that given by Hoffman and Tsai–Wu (as shown in Fig. 1).

For 2D-C/SiC composites with strong fiber/matrix interface, the reported tensile strength is $X_t=246.6$ MPa, the compressive strength is $X_c=389.6$ MPa, and the shear strength is $S_{12}=128.5$ MPa.[5] The rupture data of the material along with theoretical predictions under tension-shear combined loadings[5] are depicted in Fig. 2. It is presented in this figure that both Tsai–Hill and Hoffman criterion fail to give accurate results, because the mechanical fracture depends on the damage initiation and propagation path. This means that the damage mechanics of fibrous composites may be appropriate for characterizing these brittle failure cases.

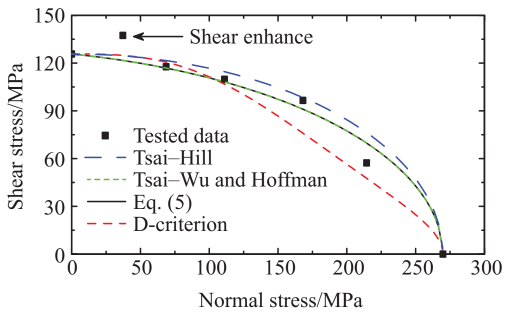


Fig. 1. Predicted failure curves by different criteria for 2D-C/SiC composites with weak interface.

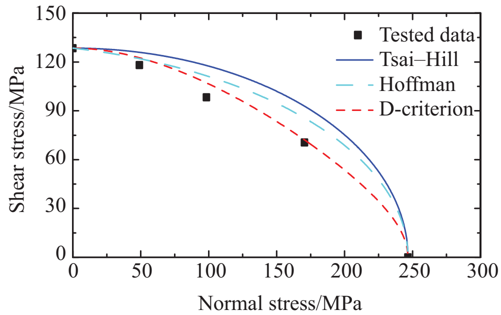
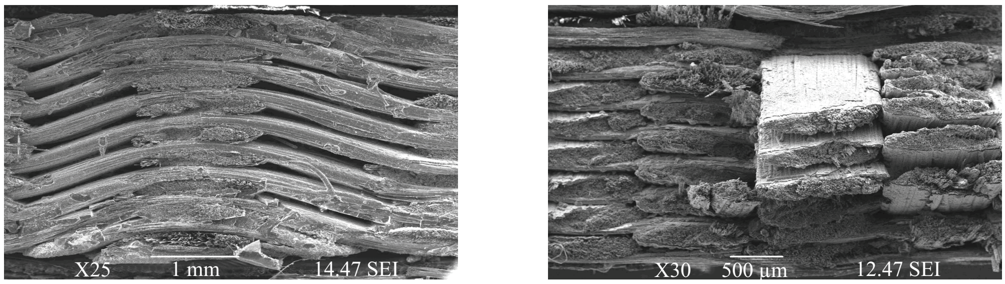


Fig. 2. Predicted failure envelops by different criteria for 2D-C/SiC composites with strong interface.

For 2.5D-C/SiC composites, experimental studies[6] suggested that the damage evolution paths and fracture modes of this highly anisotropic material are diverse and elusory under complex stress state. Therefore, the failure behavior is examined on two fracture planes, as shown in Fig. 3.

The tested results of the material under tension-shear and compression-shear combinations are described in Fig. 4. Based on the distribution regularity of the experimental data represented in Fig. 4(a), the Sun proposal[7] is accepted herein to account for the variation in shear strength due



(a) (b)

Fig. 3. Fracture plane of 2.5D-C/SiC material: (a) plane-1, (b) plane-2.

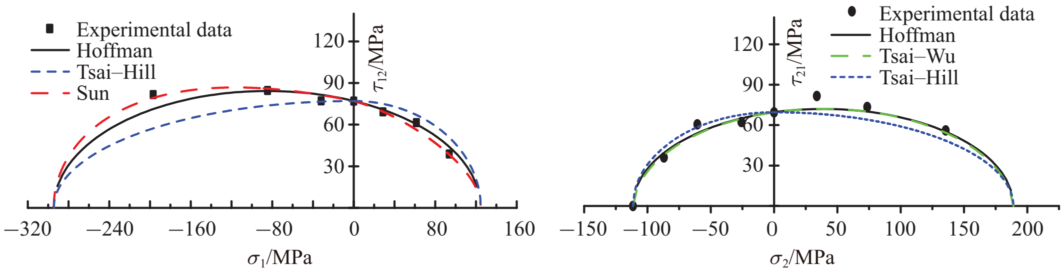
to σ_1. If tensile σ_1 is applied to the material, the shear strength is expected to decrease, such as

$$(\sigma_1 X_t)^2 + [\tau_{12}/(S_{12} - \eta_1 \sigma_1)]^2 = 1, \tag{7a}$$

where η_1 is an experimentally determined constant. Contrarily, if compressive σ_1 is imposed on the material, the shear strength will increase, that is

$$(\sigma_1/X_c)^2 + [\tau_{12}/(S_{12} - \eta_2 \sigma_1)]^2 = 1, \tag{7b}$$

where η_2 is regarded as an internal material friction parameter. For the target material, $\eta_1 = 0.2$, $\eta_2 = 0.15$. The predicted curve of Eq. (7) is plotted in Fig. 4(a), and it is in better agreement with experimental data, as compared with either the curve given by Tsai–Hill or that derived from Hoffman criteria. The failure assessment on plane-2 is described in Fig. 4(b). It is worth mentioned that the failure of the material is insensitive to a change of sign of shear stress. Thus, although the linear term of shear stress is added to Tsai–Wu criterion, the prediction has little difference with that of Hoffman.



(a) plane-1 (b) plane-2

Fig. 4. Failure strength of 2.5D-C/SiC composite under combined loadings.

In general, the phenomenological strength theories are appropriate for orthotropic materials with brittle fracture behavior. To nonlinear ceramic composites, such as C/SiC materials, the Hoffman and Tsai–Wu criteria can also provide good approximation of mechanical failure strength under certain loading conditions. However, since the classical criteria are not based on the interior damage of materials, the influence of damage mechanisms on final failure can not be accounted for. In the following analysis, a kind of damage based criterion will be proposed.

In order to describe the highly anisotropic and the direction dependent damage of nonlinear brittle matrix composites, a significant and special damage matrix is introduced herein, i.e.

$$D_{ij} = \begin{pmatrix} D_{11} & D_{12} & D_{16} \\ D_{21} & D_{22} & D_{26} \\ D_{61} & D_{62} & D_{66} \end{pmatrix}. \tag{8}$$

In the above equation, D_{ij} ($i = 1,2,\ j = 1,2,6$) denotes damage in principal material direction i caused by σ_j; while D_{6j} ($j = 1,2,6$) represents shear damage induced by σ_j. In the matrix, the six off-diagonal terms are used to characterize the damage coupling effects.

For simplicity, the damage coupling terms are all assumed to be zero in this study. The damage behavior is subsequently characterized by the residual damage components D_{11}, D_{22}, and D_{66}. It is considered that the failure of a ceramic matrix composite is driven by its interior damage. When mechanical rupture occurs, these damage components should satisfy a mathematical relationship. Luo et al.[8] had proposed a condition in which the principal maximum damage value reaches 1.0. This is similar to the maximum stress/strain criterion. However, it is also reasonable to follow the conception of classical quadratic theories, and suggest a quadratic damage function. Consequently, we introduce a reference damage variable, D_{eff}, which is quantified in mathematical perspective by a quadratic equation given as $D_{11}^2 + D_{22}^2 + D_{66}^2 = D_{\text{eff}}^2$. Generally, D_{eff} varies with loading conditions. Once mechanical failure occurs under a certain stress state, D_{eff} will achieve one of its critical values. This means that one collapse load corresponds to one reference value. In order to unify the diverse failure conditions, we propose the following damage function

$$I_{11}D_{11}^2 + I_{22}D_{22}^2 + I_{66}D_{66}^2 = 1. \tag{9}$$

In the above equation, I_{11}, I_{22}, and I_{66} are simply set as invariable, and they can be determined by some simple mechanical tests, such as uniaxial tension, compression and in-plane shear.

If one imposes a uniaxial tensile/compressive stress oriented along axis-1 or axis-2 on a specimen, then the failure conditions are

$$I_{11}\tilde{D}_{11}^2 = 1,\ \ I_{22}\tilde{D}_{22}^2 = 1, \tag{10}$$

where $\tilde{D}_{11}$ and $\tilde{D}_{22}$ are the maximum value of D_{11} and D_{22}, respectively. Similarly, by imposing pure in-plane shear stress on a sample, we obtain

$$I_{66}\tilde{D}_{66}^2 = 1, \tag{11}$$

where $\tilde{D}_{66}$ is the maximum shear damage. The solution of Eqs. (9)–(11) gives the damage-based failure criterion as

$$\left(D_{11}/\tilde{D}_{11}\right)^2 + \left(D_{22}/\tilde{D}_{22}\right)^2 + \left(D_{66}/\tilde{D}_{66}\right)^2 = 1. \tag{12}$$

Note that the basic conception of this criterion is that the final failure of composites is dominated by the interior damage degree. Equation (12) is a macroscopic proposal (referred to as D-criterion), which is prepared for advanced ceramic composites with nonlinear mechanical behavior. The shortcoming of the theory lies in the difficulty of anisotropic damage evolution analysis. Meanwhile, modifications are also essential to incorporate the damage coupling terms. Before applications, the damage evolutionary equations of target materials need specific investigation.

The unloading modulus is suggested for damage definition, which gives

$$D_{11} = 1 - E_{1\mathrm{u}}/E_1, \;\; D_{22} = 1 - E_{2\mathrm{u}}/E_2, \;\; D_{66} = 1 - G_{6\mathrm{u}}/G_6, \tag{13}$$

where E_1 and $E_{1\mathrm{u}}$ are initial modulus and unloading modulus in axis-1, E_2 and $E_{2\mathrm{u}}$ are initial modulus and unloading modulus in axis-2, and G_6 and $G_{6\mathrm{u}}$ are initial shear modulus and shear unloading modulus.

To 2D-C/SiC composites examined previously, one can derive $D_{11} = D_{22}$ for structural symmetry. According to Ref. 9, by fitting of experimental data, D_{11} (or D_{22}) can be expressed as

$$D_{11} = 3.6 \times 10^{-3}\sigma_1 - 5.7 \times 10^{-6}\sigma_1^2, \;\; 0 \leqslant \sigma_1 \leqslant 246.6\,\mathrm{MPa} \tag{14a}$$

for the 2D-C/SiC composite with strong interface. While for the material with weak interface, one can get the following relations

$$D_{11} = \begin{cases} 0, \;\; \sigma_1 < 10\,\mathrm{MPa}, \\ 0.59 - 0.73/[1+\exp(0.018\sigma_1 - 1.6)], \;\; 10\,\mathrm{MPa} \leqslant \sigma_1 \leqslant 269.8\,\mathrm{MPa} \end{cases} \tag{14b}$$

according to Ref. 10. The relationship between shear damage and shear stress is linear, so that one has

$$D_{66}/\tilde{D}_{66} = \tau_{12}/S_{12}. \tag{15}$$

By substituting the maximum value of σ_1 into Eq. (14), $\tilde{D}_{11}$ can be obtained as 0.541 for the material with strong interface and 0.563 for the material with weak interface, respectively. But the value of $\tilde{D}_{66}$ is not needed here because one can use the stress instead of the damage parameter (see Eq. (15) and Table 1), though it can be determined in the same way.

The D-criterion is preliminarily validated with respect to the off-axis loading of the 2D-C/SiC composite with weak interface. The reported off-axis strengths are 256.7 MPa ($\theta = 15°$), 203.8 MPa ($\theta = 30°$), and 195.6 MPa ($\theta = 45°$), respectively.[10] Note that θ is the off-axis angle and the stress components (σ_1, σ_2, τ_{12}) can be obtained by resolution of the off-axis strength. The predictions of D-criterion, Tsai–Hill criterion and Hoffman criterion are summarized in Table 1. Further comparison reveals that the newly proposed Eq. (12) is applicable even though it excludes the damage coupling terms.

Further applications of D-criterion are conducted for 2D-C/SiC composites under tension-shear combined loadings, and the predictions are exhibited in Figs. 1 and 2. It can be seen from the figures that the failure envelopes given by D-criterion are lower than that derived from Tsai–Hill and Hoffman criteria. To 2D-C/SiC with strong interface, the prediction of D-criterion is closer

Table 1. Evaluation results of different failure criteria.

Criterion	Theoretical function[10]	Deviation		
		15°	30°	45°
Tsai–Hill	$(\sigma_1/269.8)^2+(\sigma_2/269.8)^2-\sigma_1\sigma_2/269.8^2+(\tau_{12}/125.7)^2-1$	–0.004	–0.258	–0.263
Hoffman	$[\sigma_1^2+\sigma_2^2+(430.2-269.8)(\sigma_1+\sigma_2)]/(269.8\times430.2)-\sigma_1\sigma_2/(269.8\times430.2)+(\tau_{12}/125.7)^2-1$	–0.077	–0.069	–0.042
D-criterion	$\{\{0.59-0.73/[1+\exp(0.018\sigma_1-1.6)]\}^2+\{0.59-0.73/[1+\exp(0.018\sigma_2-1.6)]\}^2\}/0.563^2+(\tau_{12}/120)^2-1,\ \ (\sigma_1\geqslant 10\ \mathrm{MPa}, \sigma_2\geqslant 10\ \mathrm{MPa})$	+0.197	+0.070	+0.013

to the experimental results for the stochastic damage initiation and brittle fracture. However, the new criterion is not used to the 2.5D-C/SiC composite, because the damage evolution equations are not available in the literature.[6]

The present paper has to be considered as a preliminary investigation. The applicability and limitations of the classical phenomenological strength theories to C/SiC composites are investigated. A kind of damage-based criterion named D-criterion is proposed for advanced ceramic composites. The preliminary validation is performed under off-axis tension and tension-shear combined loadings. The prediction shows that the failure envelope given by D-criterion is governed by damage evolution equations, and more complex failure envelops can be described besides quadratic curves. Future work will concentrate on improving the D-criterion in order to account for the influence of damage coupling on final failure of brittle matrix composites.

This work was supported by the Basic Research Funds of Northwestern Polytechnical University (JC20110219) and the National Natural Science Foundation of China (11102160).

1. V. D. Azzi, S. W. Tsai. Anisotropic strength of composites. *Experimental Mechanics* **5**, 283–288 (1965).
2. O. Hoffman. The brittle strength of orthotropic materials. *Journal of Composite Materials* **1**, 200–206 (1967).
3. S. W. Tsai, E. M. Wu. A general theory of strength for anisotropic materials. *Journal of Composite Materials* **5**, 58–80 (1971).
4. G. Y. Guan, G. Q. Jiao, Z. G. Zhang. Phenomenal failure criteria of a plain woven C/SiC composite. *Journal of the Chinese Ceramic Society* **33**, 1100–1104 (2005).
5. X. Wang, B. Wang, G. Q. Jiao. Experimental investigation on plain woven C/SiC composites under complex stress states. *Journal of Mechanical Strength* **32**, 35–39 (2010).
6. S. F. Liang. Mechanical behavior of self-healing 2.5D-C/SiC composite, [PhD Thesis]. Northwestern Polytechnical University, Xi'an (2013) (in Chinese).
7. C. G. Dávila, P. P. Camanho, C. A. Rose. Failure criteria for FRP laminates. *Journal of Composite Materials* **39**, 323–345 (2005).
8. D. Luo, S. Takezono, K. Tao, et al. The mechanical behavior analysis of CFCC with overall anisotropic damage by the micro-macro scale method. *International Journal of Damage Mechanics* **12**, 141–162 (2003).
9. C. P. Yang, G. Q. Jiao, B. Wang. Uniaxial tensile stress-strain behavior and strength of plain woven C/SiC composite. *Chinese Journal of Theoretical and Applied Mechanics* **43**, 330–337 (2011) (in Chinese).
10. G. Y. Guan. Study on mechanical behavior of plain woven C/SiC ceramic matrix composites, [PhD Thesis]. Northwestern Polytechnical University, Xi'an (2005) (in Chinese).